Rank-27 Cluster Timing Stability

Tactix Development Project

26 August 2026

Executive summary

The 300–580 second Rank-27 spread is execution-environment noise, not solver nondeterminism, and Kissat exhibits the same effect. Immutable runs perform the same logical work and essentially the same instructions, but solvers sharing one L3 domain consume materially more cycles at lower IPC. An exact contemporary Tactix control needs 367.312 seconds at the isolated median and 419.659 seconds when only one other Rank-27 solver shares the node. Instructions change 0.025%, while cycles rise 14.231% and IPC falls 1.805 to 1.580. Shared cache and memory service rate is the owner; CPU descheduling, migrations, average frequency, and proofless I/O are not.

The completed node-isolation experiment is causal evidence, not a future timing or diagnostic policy on a two-node cluster. Seven isolated proof-off observations span 365.171–371.637 seconds; three primary repeats have median 367.312 and 1.237% range. Three isolated proof-on observations have median 370.893 and 0.437% range. More importantly for future policy, six ordinary shared-node `-c 12` controls, each reserving one complete L3 domain, reproduce the accepted Tactix trajectory in 367.045–372.354 internal seconds (median 368.307, full range 1.441%). Every run is below 400 seconds.

Proof production itself costs only 3.581 seconds at the isolated median (0.975%), not the historical unstable estimate of 28.532 seconds. Proof checking is separate: median 112.03 seconds. All proof jobs reproduce the current candidate's 1,514,395,111-byte proof and pass pinned `drat-trim`. I/O is not the present solver bottleneck because proofs and inputs reside on node-local NVMe; the repository is NFS but receives only compact evidence.

The approved policy uses ordinary shared-node `-c 12` allocations with exactly 48,000 MiB of scheduler memory while retaining the 16-GiB solver process limit, simultaneous separate candidate/baseline jobs, three independent paired waves for decision-grade performance claims, and median ratio/full-spread reporting. Fuzzing remains one Claude-owned campaign and is not repeated by this policy. Future whole-node isolation is prohibited.

1 Evidence and cause

The immutable binary pin is SHA-256 `4b3f7467...bc5e6a` from pushed commit `d8da6a1`. The input SHA-256 is `5435b1b2...f7bb4e9`. Raw authority is the 10:30 Rank-27 timing-stability campaign of 26 August; its summary SHA-256 is `99c43cf1...abab0b`. The manifest records the full campaign name and immutable identities.

Historical SLURM accounting was reconstructed across every partition sharing node017 and node018. The exact `60fe8e` binary takes 553.948 seconds on node017 while overlapping 79.50 other allocated CPUs, but 394.107 seconds on idle node018. The exact BVE pin takes 419.772 seconds with 60.92 other CPUs on node017 and 371.711, 375.797, and 372.969 seconds across three idle-node018 runs. The original 399.152-second witness itself ran on idle node017, so the node label is not causal. The measured

slow intervals used `cpu-dense-hwulab-q`; no selected interval had a preempt-partition co-tenant, but both partitions use the same physical nodes and must be audited.

Frozen Kissat 4.0.4 supplies an independent exact-binary reproduction. On the same immutable input it takes 278.187 seconds in a `-c 4` job and 320.307 seconds in a `-c 2` job, a 15.14% spread. Both runs generate the same 1,320,603,112-byte proof at SHA-256 `ca29c36...d6d3f`; both pass the independent DRAT-to-Cake pipeline and leave zero residue. The faster run actually overlaps more allocated CPUs (88.00 versus 73.90), proving that raw node occupancy is not a sufficient predictor.

All twelve current jobs complete 0:0, each with one solver invocation. There is no node constraint or task throttle. Every solver returns UNSAT exit 20. All proof-off runs share normalized no-timing SHA-256 `10372d94d2cb1ae86893408435e0c3ffa2338053b11b7012c2107a3bebf1ffd0`; all proof-on runs share their mode-specific SHA-256 `4cfab3ac1e0041219c6d5a5a201d9672d4f2b5983a15705e373b674e998df3f0`. Residue is zero.

Proof-off condition	Internal s	Cycles	Instructions	IPC
Isolated median (three)	367.312	1.390×10^{12}	2.509×10^{12}	1.805
Two simultaneous, node017	419.659	1.588×10^{12}	2.509×10^{12}	1.580

The shared run adds 52.347 seconds and 14.231% cycles with only 0.025% additional instructions. Both have one CPU migration and approximately 3.781-GHz average frequency. The modest cache-miss-count change coupled with the large IPC loss is consistent with longer shared-miss and bandwidth service, not changed search, frequency collapse, or loss of scheduled CPU time.

2 Kissat reproduction and shared-L3 remedy

Campaign `20260826-1430-rank27-kissat-c4-timing-variation` runs 15 proofless controls from exact instrumented Kissat SHA-256 `b6a9b1d0...644d5`. All return UNSAT exit 20 and reproduce 4,120,691 conflicts, 171,577,295 decisions, 2,665,923,683 propagations, and every selected inprocessing counter. Inputs are staged on local NVMe; cleanup and residue are zero.

Kissat protocol	Median s	Min–max s	Range / median
Six simultaneous `-c 4`	298.11	253.90–321.32	22.62%
Three separated `-c 4`	260.02	258.60–260.12	0.585%
Six simultaneous `-c 12`	259.13	252.75–260.76	3.09%

The topology probe explains the packed `-c 4` failure. One EPYC 9454 L3 domain covers six physical cores. On lightly allocated node018, solvers pinned to cores 0, 2, and 4 share one L3 and take 297.74–300.11 seconds at IPC 1.825–1.841; simultaneous core 6 lands in the next L3 and takes 259.66 seconds at IPC 2.110. A separated core-0 repeat recovers to 258.60 seconds and IPC 2.120. Instructions remain about 2.071×10^{12} while IPC spans 1.706–2.159: exact work receives different cache/memory service.

Ordinary `-c 12` masks such as `0-5,96-101` reserve exactly one L3 domain without reserving a node. The nodes contain 16 such domains, permitting 16 simultaneous timed solvers per node. Six accepted-Tactix controls under the same policy reproduce normalized no-timing SHA-256 `10372d94...f1ffd0`. Their internal times are 367.045, 372.354, 368.016, 368.450, 368.164, and 368.731 seconds: median 368.307, full range 5.309 seconds or 1.441%. All are below 400 seconds and residue-free. This is the measured shared-node remedy.

3 Diagnostic calibration and CPU allocation

Bare `-exclusive` is unavailable on this partition. The supported combination `-exclusive=user -mem=0` provides effective isolation: exclusive-user ownership excludes other users across partitions, and reserving all memory prevents another same-user job fitting. A shell `ulimit` independently retains the actual 16-GiB solver contract. Node snapshots show one timing job, node owner `hwu`, and all 751 GiB reserved.

Mode and protocol	Observations (s)	Median	Range
Isolated proof-off	367.092, 371.637, 367.312	367.312	4.545 (1.237%)
Isolated proof-on	370.893, 369.785, 371.407	370.893	1.622 (0.437%)
All isolated proof-off variants	365.171–371.637 (seven)	367.295	6.466 (1.761%)

The EPYC 9454 nodes have two hardware threads per core. Ordinary `-c 2` exposes `0,96`, one physical core; ordinary `-c 4` exposes `0-1,96-97`, two cores. On this cluster, `-c 2` plus `-hint=nomultithread` also exposes the two-core mask. Isolated ordinary `-c 2` runs are 366.609 and 365.171 seconds; isolated ordinary `-c 4` runs are 368.081 and 367.295. Their medians differ 0.49%, which is immaterial. Tactix is single-threaded: `-c 4` is a clearer ordinary allocation, not computational parallelism. It reserves two physical cores and matches the partition's 16-GiB-per-two-core convention, but does not reserve shared cache or memory bandwidth. It is suitable for functional work but not long-instance timing. `-c 12` is the minimum scheduler-visible allocation that reserves a complete six-core L3 domain on these nodes. It retains the 16-GiB process contract and ordinary shared scheduling.

4 Proof and I/O decomposition

Proof-on median minus proof-off median is 3.581 seconds, 0.975%. Median cycles rise 0.980% and instructions 1.336%. Every proof has 1,514,395,111 bytes and SHA-256 `7bcb3d8db7fe5827daa6af18281684b1f82f79c4f28468046a88b2ab326ac47e`; pinned `drat-trim` accepts all four. Isolated checker walls are 112.12, 112.03, and 107.11 seconds, median 112.03. The median proof-on solver plus checker pipeline is 482.923 seconds; checker time is validation overhead, not solver score.

Scratch is local NVMe XFS, while the repository is NFS over RDMA. The staged CNF, proof, and checker input remain local. Proofless output is negligible. Writing the 1.514-GB proof adds about one second of system time and negligible I/O PSI. Checking sees cached pages and reports no physical filesystem input. Thus current solver timing is not I/O-bound. Local staging remains mandatory; putting proof or CNF traffic on NFS would create a different protocol.

5 Approved policy and disposition

The user-approved stable contract is:

1. Routine timing uses separate one-solver candidate/baseline jobs, `-c 12`, an exact 48,000-MiB SLURM allocation, no whole-node exclusivity or all-memory reservation, and the applicable 16-GiB solver process limit. The allocation matches six physical cores at the effective 8,000 MiB/core; it does not raise the solver limit or reserve memory bandwidth. The exact MiB spelling permits 16 jobs within configured node memory 769,024 MiB; `48G` would round up and permit only 15. Require each CPU mask to cover one complete L3 domain. Launch both arms in the same wave as separate jobs; never serialize them in one allocation. `-c 4` remains valid for functional work, not timing.

2. Bind each solver to one granted hardware thread and record occupancy across all partitions. Occupancy is a covariate and rejection diagnostic, not a reason to reserve an otherwise idle 96-core node.

3. Require verified node-local scratch for CNF, proof, and checker work. Record node, CPU mask, time, cycles/instructions/IPC/migrations, load/PSI, proof identity, checker time, and cleanup.

4. For a performance claim that accepts or rejects a repair, require three independent exact-state paired waves. Each wave concurrently runs one separate candidate job and one separate baseline job, crossing arm placement over node/L3 positions when practical. Report the median paired candidate/baseline ratio and its full range, never best-of. An absolute headline is secondary and allowed only when both arms' complete ranges are at most 2%; otherwise it is environmentally invalid, not a solver regression. This requirement does not repeat fuzzing: each fuzz checkpoint remains one Claude-owned campaign. Correctness gates and ordinary diagnostic probes keep their existing predeclared contracts.

5. Keep proof-on solver time, proof-off diagnostic time, and checker time as three separate quantities. Never subtract unmatched shared-node runs to price proof generation.

6. Never request `-exclusive=user -mem=0` or another whole-node reservation for timing or diagnosis. Use paired ratios, cross-partition occupancy, perf service-rate evidence, and naturally observed load windows. Full-width tier arrays retain their existing eligible-width/no-node-constraint contract and use aggregate paired evidence.

Allocation-only probe `1648628` verifies `SLURM_CPUS_PER_TASK=12`, `SLURM_MEM_PER_NODE=48000`, and CPU mask `0-5,96-101`, exactly one L3 domain. Its wrapper exits nonzero only after recording that evidence because this cgroup-v1 cluster lacks the optional cgroup-v2 `memory.max` path; no solver ran. Initial probe `1648627` is infrastructure-invalid because its log was placed on compute-local scratch. Neither probe is solver or fuzz evidence.

For the accepted pin, 367.045–372.354 internal seconds (median 368.307) is the supported ordinary shared-node `-c 12` calibration. Routine development decisions use paired ratios. A stable paired slowdown is real solver evidence even if both arms exceed 400; an unstable absolute observation is not. The technical report and campaign are complete, and the approved contract is now reflected in the Development Plan. Phase 4.25 broad search-quality convergence resumes after the pending consolidation fuzz verdict and wrap-up.